Tutorial - 2

Q1 Compute the following limits:

(i) $\lim_{x\to 0} \frac{\sin(x)}{x}$ (ii) $\lim_{x\to 1} \frac{x^n - 1}{x-1}$

(i) $\lim_{x\to 0} \frac{\sin x}{x} = \lim_{x\to 0} \frac{\frac{d}{dx}(\sin(x))}{\frac{d}{dx}(x)}$ [It's $\frac{0}{0}$ form; Using L'Hôpital rules]

$= \lim_{x\to 0} \frac{\cos x}{1} = 1 \Rightarrow \boxed{\lim_{x\to 0} \frac{\sin x}{x} = 1}$

(ii) $\lim_{x\to 1} \frac{x^n - 1}{x-1} = \lim_{x\to 1} \frac{\frac{d}{dx}(x^n - 1)}{\frac{d}{dx}(x-1)} = \lim_{x\to 1} \frac{n x^{n-1}}{1}$ | $\frac{0}{0}$ form.

$\Rightarrow \boxed{\lim_{x\to 1} \frac{x^n - 1}{x-1} = n}$

Q2 If $h(x) = f(x)\, g(x)$ then show $\frac{dh(x)}{dx} = \frac{df(x)}{dx} g(x) + f(x) \frac{dg(x)}{dx}$

From the definition,

given $h(x) = f(x)\, g(x)$

$$\frac{dh(x)}{dx} = \lim_{h\to 0} \frac{h(x+h) - h(x)}{h}$$

$$= \lim_{h\to 0} \frac{f(x+h)\, g(x+h) - f(x)\, g(x)}{h}$$

$$= \lim_{h\to 0} \frac{f(x+h)\, g(x+h) - f(x)\, g(x+h) + f(x)\, g(x+h) - f(x)\, g(x)}{h}$$

$$= \lim_{h\to 0} \left[\frac{f(x+h) - f(x)}{h}\right] g(x+h) + \lim_{h\to 0} f(x) \left[\frac{g(x+h) - g(x)}{h}\right]$$

$$= \left[\lim_{h\to 0} \frac{f(x+h) - f(x)}{h}\right]\left[\lim_{h\to 0} g(x+h)\right] + f(x)\left[\lim_{h\to 0} \frac{g(x+h) - g(x)}{h}\right]$$

$$\boxed{\frac{d}{dx} h(x) = \frac{df(x)}{dx} g(x) + f(x) \frac{dg(x)}{dx}}$$

Q3 Find the maximum and minimum values of the curve $y(x) = x^3 - 3x$. Plot the function $y(x)$.

At the extrema of the curve : $\frac{dy(x)}{dx} = y'(x) = 3x^2 - 3 = 0$

$\Rightarrow \quad 3x^2 - 3 = 0 \Rightarrow \boxed{x = \pm 1}$

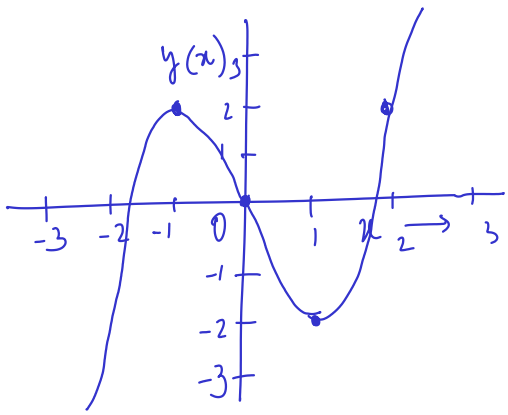
Now, 2nd derivative of the function $y''(x) = \frac{d^2 y(x)}{dx} = 6x$

(a) At $x = +1 \Rightarrow y''(x) = 6 \times 1 = 6 > 0$

$\longrightarrow$ $x = +1$ is the minimum of the curve

(b) $x = -1 \Rightarrow y''(x) = 6 \times (-1) = -6 < 0$

$\longrightarrow$ $x = -1$ is the maximum of the curve



$y(x) = x^3 - 3x$

x	y
0	0
1	-2
-1	2
2	2
-2	-2
3	

$y(-x) = (-x)^3 - 3(-x)$
$= -(3x^3 - 3x)$
$= -y(x)$

$\rightarrow$ Odd function